Name
Date
Course

Solving Quadratic Equations with the Quadratic Formula

Solve by completing the square. **Answers**

1. $0=-2x^2+5x-5$ $\boxed{\frac{5\pm i\sqrt{15}}{4}}$

2. $-4x^2+x-2=0$ $\boxed{\frac{1\pm i\sqrt{31}}{8}}$

3. $0=4x^2+2x+1$ $\boxed{\frac{-1\pm i\sqrt{3}}{4}}$

4. $3x^2+6x+5=0$ $\boxed{\frac{-3\pm i\sqrt{6}}{3}}$

5. $8x^2-4x+1=0$ $\boxed{\frac{1\pm i}{4}}$

Solve using the quadratic formula.

6. $x^2-5x+2=0$ $\boxed{\frac{5\pm\sqrt{17}}{2}}$

7. $0=-4x^2+3x+3$ $\boxed{\frac{3\pm\sqrt{57}}{8}}$

8. $2x^2+2x-3=0$ $\boxed{\frac{-1\pm\sqrt{7}}{2}}$

9. $x^2-7x+1=0$ $\boxed{\frac{7\pm3\sqrt{5}}{2}}$

* Calculator → 10. $3x^2-13x+14=0$ $\boxed{\frac{7}{3}\text{ or }2}$

11. $0=25x^2-10x+1$ $\boxed{x=\frac{1}{5}}$

12. $6x^2+x+3=0$ $\boxed{\frac{-1\pm i\sqrt{71}}{12}}$

13. $x^2-2x+5=0$ $\boxed{1\pm2i}$

14. $0=-8x^2+4x-3$ $\boxed{\frac{1\pm i\sqrt{5}}{4}}$

15. $x^2+3x+1=0$ $\boxed{\frac{-3\pm\sqrt{5}}{2}}$

16. $-3x^2+9x-3=0$ $\boxed{\frac{3\pm\sqrt{5}}{2}}$

17. $10x^2+11x+3=0$ $\boxed{-\frac{3}{5}\text{ or }-\frac{1}{2}}$

18. $x^2-x+16=0$ $\boxed{\frac{1\pm3i\sqrt{7}}{2}}$

Find the discriminant of each of the following quadratic equations and use it to determine the number and type of solutions. Write "one real solution," "two real solutions," or "two complex solutions."

19. $6x^2-2x+1=0$ $\boxed{D=-20}$ Two complex solutions

20. $0=x^2-x-4$ $\boxed{D=17}$ Two real solutions

(21) $10x^2+5x-3=0$

$\boxed{\begin{array}{c} D=145 \\ \text{Two real} \\ \text{Solutions} \end{array}}$

(22) $0=4x^2+20x+25$

$\boxed{\begin{array}{c} D=0 \\ \text{One real} \\ \text{Solution} \end{array}}$

(23) $0=-x^2+x-1$

$\boxed{\begin{array}{c} D=-3 \\ \text{Two Complex} \\ \text{Solutions} \end{array}}$

(24) $16x^2-8x+1=0$

$\boxed{\begin{array}{c} D=0 \\ \text{One real} \\ \text{Solution} \end{array}}$